

Data-Driven Vehicle Longevity

Using computing and engineering judgment to understand which vehicle designs tend to stay useful longer.

47,548

EPA vehicle model records analyzed

1985-2024

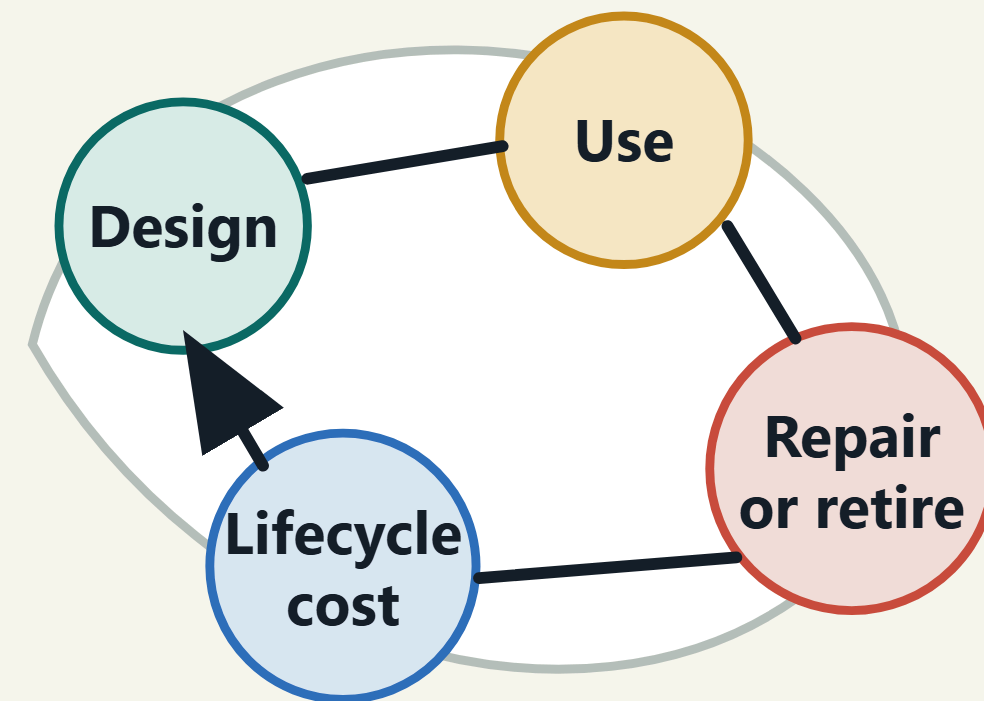
model years represented in the dataset

5

fuel groups compared for lifespan patterns

Longevity changes the real cost of a vehicle.

- Vehicles that last longer can reduce replacement demand and ownership cost.
- Longer lifespan can also extend emissions from older technology.
- Engineers need evidence to balance durability, efficiency, and environmental impact.



I studied how long vehicles stay in use.

- **What I measured:** estimated years a vehicle remains on the road.
- **What I compared:** fuel type, engine size, and efficiency.
- **What made it tricky:** newer cars have not had enough time to show their full lifespan.

Older model



We can estimate when it likely left the road.

Newer model



It may still be operating, so the story is incomplete.

The model lets both kinds of information count without pretending the newer vehicles are finished.

METHOD

From EPA records to engineering insight.

1

Collect

EPA Fuel Economy data covering vehicle models from 1985 through 2024.

2

Prepare

Group vehicles by fuel type and estimate vehicle age from model year.

3

Compare

Estimate which vehicle groups tend to stay on the road longer.

4

Translate

Turn patterns into design, cost, and lifecycle decisions.

Code

Cleaned and grouped thousands of records in R.

Charts

Made lifespan patterns visible for non-specialists.

Decisions

Connected results to durability and replacement planning.

MAIN FINDING

Fuel type was a strong predictor of vehicle longevity.

- Diesel vehicles showed the strongest long-term pattern in the lifespan curve.
- Midgrade gasoline vehicles had the shortest median lifespan in the dataset.
- Regular gasoline had the largest evidence base, with 30,374 records.

The plot shows that the vehicle groups separate over time instead of following one shared lifespan pattern.

30,374

regular-gas records, the largest group

7 yrs

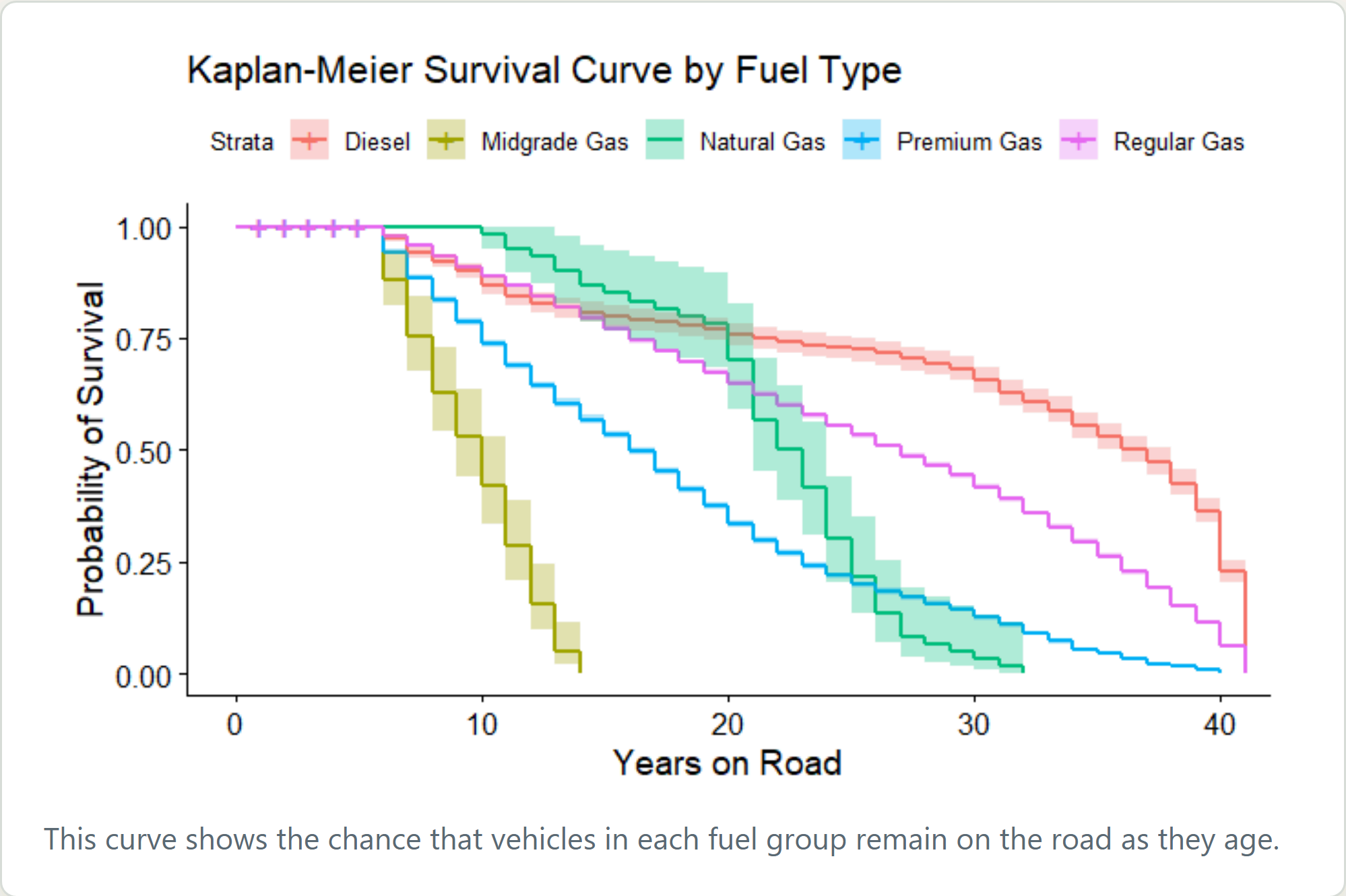
midgrade median lifespan in the dataset

25 yrs

regular-gas median lifespan in the dataset

41 yrs

maximum lifespan in analysis



The finished analysis creates evidence people can act on.



Fleet planning

Use lifespan patterns to plan replacements before maintenance costs become unpredictable.



Design tradeoffs

Compare durability, fuel efficiency, warranty risk, and long-term system performance.



Future analysis

Add mileage and maintenance data next to make the model stronger and more predictive.

30,374

regular-gas records anchor the largest comparison group

37.8%

midgrade records are newer/incomplete, so uncertainty matters

14.1 yrs

diesel lifespan varies widely, which raises design questions

Next

connect lifespan estimates with mileage, repairs, and retirement records

I connected coding work to engineering judgment.

I cleaned the data

I turned a large EPA dataset into variables that answered a lifespan question.

I built comparisons

I used R to compare fuel groups and show how vehicle age patterns changed over time.

I interpreted results

I connected the output to durability, lifecycle cost, replacement planning, and emissions tradeoffs.

I translated the story

I made the technical work understandable for engineering peers, managers, and non-specialists.

This is the kind of engineer I am working to become.

Mechanical foundation

I care about how real systems wear, fail, and keep delivering value.



Computing toolkit

I use data, code, and visualization to make system behavior measurable.



Engineering impact

I want to improve predictive maintenance, lifecycle planning, and durable design.

This project showed me how computing can help mechanical engineers make better decisions before systems reach the end of their useful life.

CONCLUSION

Key takeaway: lifespan is a design decision, not just an age number.

- **Does fuel type cause lifespan?** It is connected, but maintenance, usage, and design also matter.
- **What would improve the project?** Add mileage, repair history, and actual retirement records.
- **Why does this fit my goals?** It uses code to make mechanical system performance easier to predict.

